Experiment No. 1

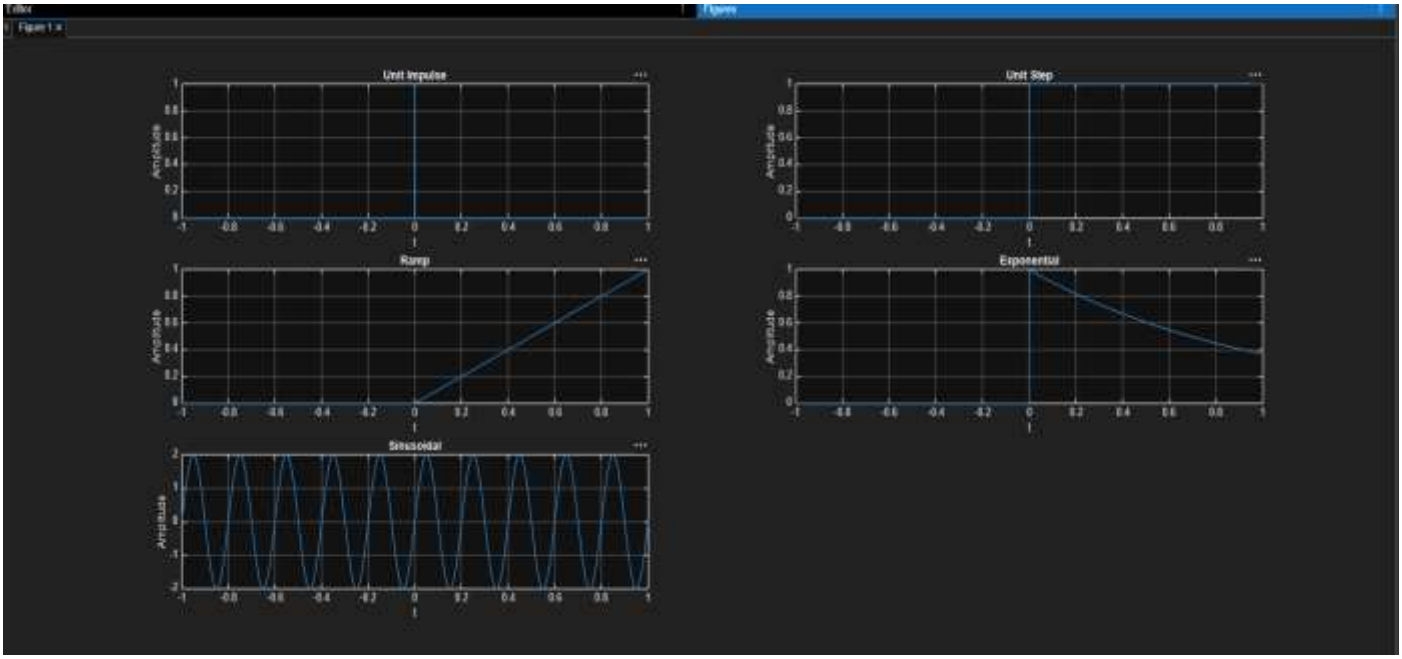
Generation and Classification of Continuous-Time and Discrete-Time Signals

(i)

/MATLAB Drive/Experiment_1_1.m

```
clc;
clear;
close all;
t = -1:0.001:1;
A = 2; f0 = 5; a = 1;
imp = double(abs(t)<0.0005);
step = double(t>=0);
ramp = t.*step;
expSig = exp(-a*t).*step;
sinSig = A*sin(2*pi*f0*t);
figure;
subplot(3,2,1); plot(t,imp); grid on; title('Unit Impulse'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,2); plot(t,step); grid on; title('Unit Step'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,3); plot(t,ramp); grid on; title('Ramp'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,4); plot(t,expSig); grid on; title('Exponential'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,5); plot(t,sinSig); grid on; title('Sinusoidal'); xlabel('t'); ylabel('Amplitude')
```

Code



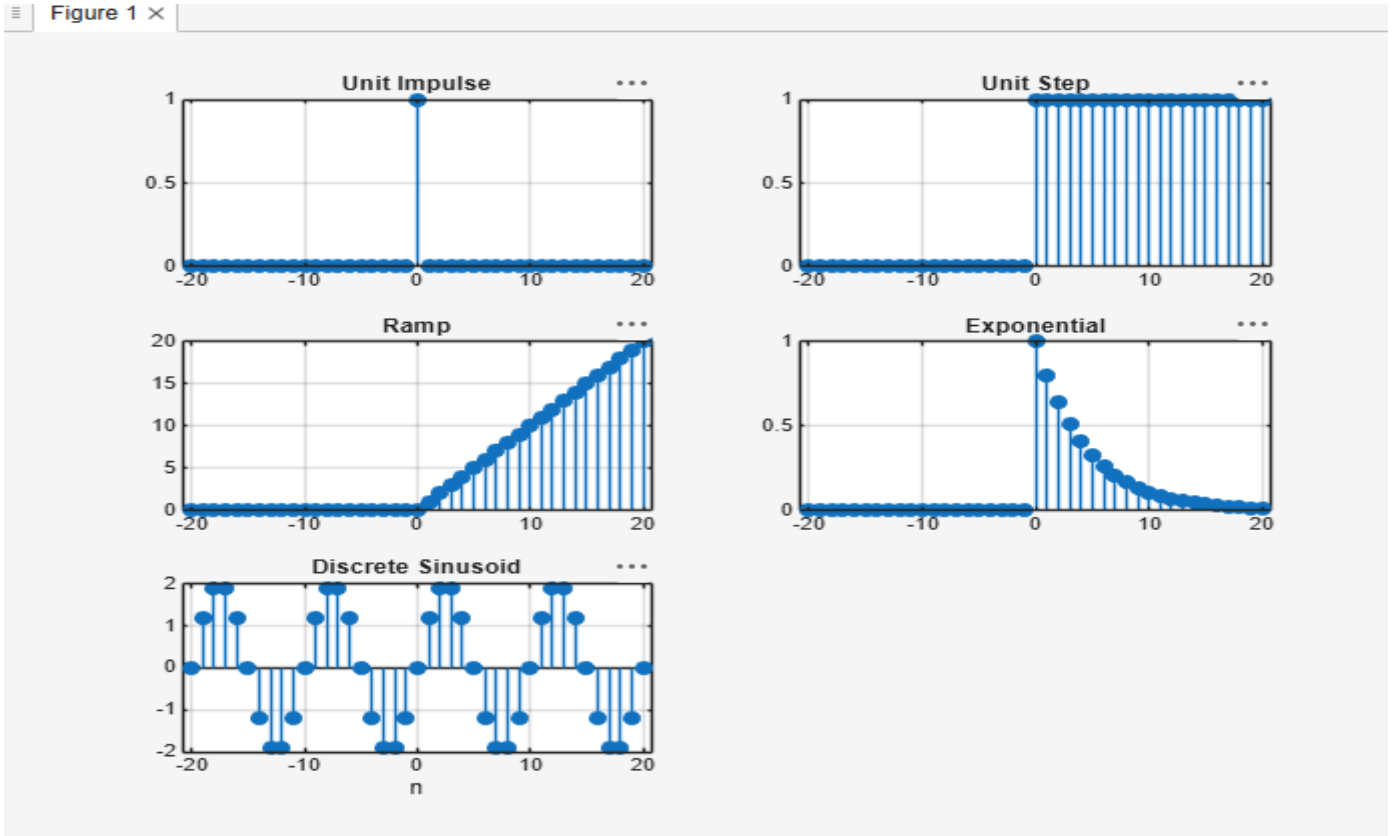
Output

(ii)

/MATLAB Drive/Experiment_1_2.m

```
1   clc; clear; close all;
2   n = -20:20;
3   A = 2; f0 = 0.1; a = 0.1;
4   imp = double(n==0);
5   step = double(n>=0);
6   ramp = n.*step;
7   expSig = (0.8).^n .* step;
8   sinSig = A*sin(2*pi*f0*n);
9   figure;
10  subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
11  subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
12  subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
13  subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
14  subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinusoid');
15  xlabel('n');
```

Code



Output